



WHD Series

Heatless Desiccant Air Dryers

Where a +3 °C dew point is not dry enough, WHD heatless dryers pass compressed air through a bed of activated alumina or molecular sieve. One tower dries while the other regenerates with a small flow of purge air, switching every few minutes.

Capacity

300 to 2000 CFM

Pressure dew point

**–20 °C (alumina) · –40 °C
(molecular sieve)**

Working pressure

7 to 12.5 bar g

Inlet temperature

45 °C

Win Equipments

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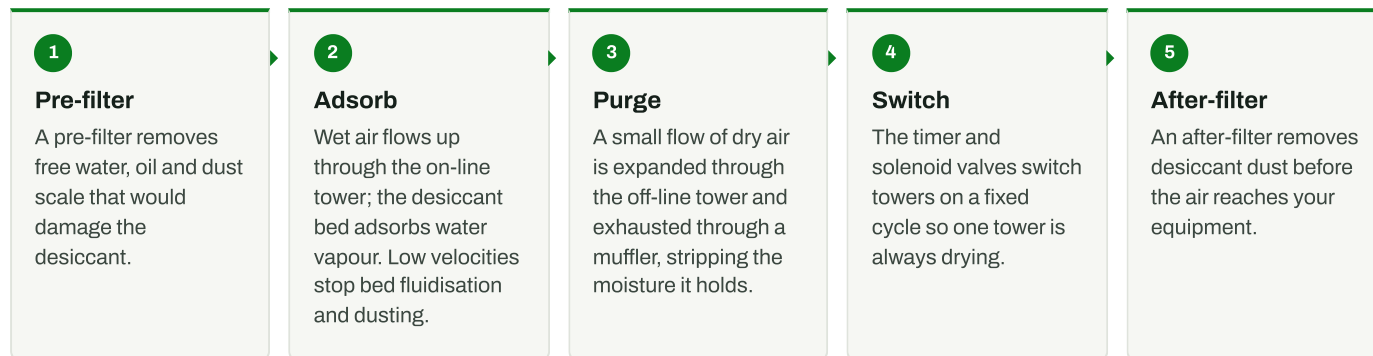
Full model range,
photos and
sizing tools



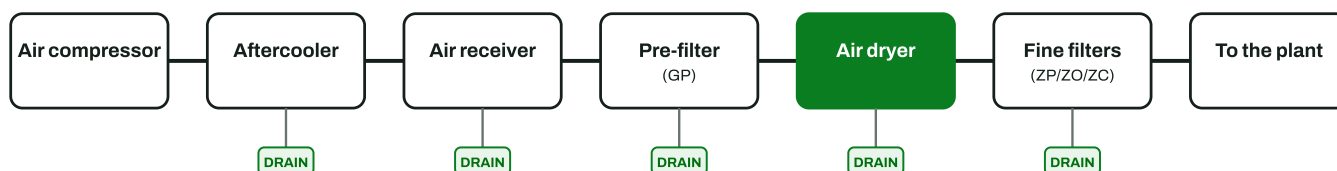
WORKING PRINCIPLE

How it works

A heatless desiccant dryer removes water vapour by adsorption onto a desiccant bed. No heater is used: a small part of the dried air regenerates the off-line tower.



TYPICAL INSTALLATION



Automatic drain valves discharge condensate from the aftercooler, receiver, filters and dryer.

CONSTRUCTION

Features

Controls

- Power-on light and on-off switch
- Accurate motorised cam timer
- PLC and cycle-failure indication (optional)

Frame

- Structural steel frame with floor stand
- Lifting lugs
- Optional factory-mounted pre- and after-filters

Desiccant bed

- Large bed for sufficient contact time
- Sized to save heat of adsorption, minimising purge air
- Bulk desiccant: no special cartridges

Quiet running

- Heavy-duty purge exhaust mufflers

Valves

- Non-lubricated inlet control valves
- Purge flow valve to adjust purge rate
- Pressure relief valve on each tower

Options

- Higher pressures and lower dew points on request

WHD Series models

Rating conditions

Dew point rating At 38 °C inlet and 7 bar g	Design pressure 13.2 bar g	Cycle time 10 min	Power supply 220 V AC ± 5%, 1 Ph, 50 Hz
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Model	Flow rate CFM	Flow rate m³/hr	Desiccant options	Electrical V / Ph / Hz	L × W × H mm	In / out NB	Weight kg
WHD 030	300	510	A / M	220 / 1 / 50	1450 × 1100 × 2500	1½"	450
WHD 040	400	680	A / M	220 / 1 / 50	1450 × 1200 × 2500	2"	630
WHD 050	500	850	A / M	220 / 1 / 50	1650 × 1250 × 2300	3"	720
WHD 060	600	1020	A / M	220 / 1 / 50	1650 × 1250 × 2500	3"	910
WHD 075	750	1275	A / M	220 / 1 / 50	2100 × 1400 × 2300	4"	1100
WHD 100	1000	1700	A / M	220 / 1 / 50	2100 × 1400 × 2500	4"	1200
WHD 125	1250	2125	A / M	220 / 1 / 50	2100 × 1400 × 2800	6"	1500
WHD 150	1500	2550	A / M	220 / 1 / 50	2300 × 1500 × 2300	6"	2100
WHD 200	2000	3400	A / M	220 / 1 / 50	2300 × 1500 × 2700	6"	2400

- › A: activated alumina, pressure dew point –20 °C at 38 °C and 7 bar g working pressure.
- › M: molecular sieves, pressure dew point –40 °C at 38 °C and 7 bar g working pressure.
- › Design pressure 13.2 bar g. Standard supply 220 V AC ± 5%, 1 Ph, 50 Hz.
- › Model code: WHD + capacity code (e.g. WHD 100 = 1000 CFM) + desiccant type.

APPLICATIONS

Pharmaceutical

Electronics

Instrument air

Food packaging

Paint lines

Outdoor air lines in cold conditions

FROM OUR WORKS



Heatless desiccant air dryer



Compact twin-tower desiccant dryer

QUESTIONS

Alumina or molecular sieve: which do I need?

Activated alumina gives a -20°C pressure dew point; molecular sieve gives -40°C (both at 38°C inlet and 7 bar g). Choose molecular sieve when your process needs the lower dew point.

What does heatless mean?

The off-line tower is regenerated with a small flow of dry purge air instead of a heater, so there is no heater to maintain.

Why does a desiccant dryer need filters?

A pre-filter keeps free water and dust off the desiccant, a zero-oil filter stops compressor oil from coating it, and an after-filter catches desiccant dust before it reaches your equipment.

What working pressure is supported?

Standard WHD dryers work from 7 to 12.5 bar g with a design pressure of 13.2 bar g. Higher pressures can be offered on request.

Talk to an engineer

Send your compressor capacity, heat load or water flow, operating pressure, temperatures and site conditions. We reply with a recommended model and quotation.

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ISO 9001:2015 certified. Specifications are as per our catalogue and are subject to change without notice; confirm the selection with our engineers before ordering. Issued October 2026.